

Black's Zero-Beta Asset Pricing Model

Capital Market Equilibrium with Restricted Borrowing

1 Overview

The central contribution of Black's zero-beta asset-pricing model is that the standard linear relation between expected return and market beta can survive even when investors are unable to borrow freely at the risk-free rate.

The standard Sharpe–Lintner CAPM is

$$E(R_i) = R_f + \beta_i [E(R_M) - R_f], \quad (1)$$

where

$$\beta_i = \frac{\text{Cov}(R_i, R_M)}{\text{Var}(R_M)}. \quad (2)$$

A key assumption behind this version of the CAPM is that investors are able to borrow and lend unlimited amounts at the same risk-free rate R_f .

Black shows that this assumption is not necessary for obtaining a linear beta–expected-return relation.

2 The Problem with Unrestricted Risk-Free Borrowing

In the standard CAPM, an investor who wants to hold a portfolio with a beta greater than one can borrow at the risk-free rate and invest more than 100 percent of wealth in the market portfolio.

For example,

$$R_p = (1 - w)R_f + wR_M, \quad w > 1. \quad (3)$$

The beta of this portfolio is

$$\beta_p = w. \quad (4)$$

Therefore, if $w = 1.5$, the investor effectively holds

150% in the market portfolio

and finances the additional investment by borrowing

50% at the risk-free rate.

In reality, however, individual investors generally cannot borrow unlimited amounts at the Treasury-bill rate. Black therefore studies capital-market equilibrium when borrowing at the risk-free rate is restricted or unavailable.

3 Black's Main Result

Black shows that expected returns remain linear in beta even without unrestricted risk-free borrowing.

The pricing equation becomes

$$\boxed{E(R_i) = E(R_Z) + \beta_i [E(R_M) - E(R_Z)]} \quad (5)$$

where R_Z denotes the return on the **zero-beta portfolio**.

This equation is the central result of the Black zero-beta CAPM.

The main difference from the Sharpe–Lintner CAPM is therefore

$$\boxed{R_f \longrightarrow E(R_Z)} \quad (6)$$

in the intercept of the Security Market Line.

4 What Is the Zero-Beta Portfolio?

The zero-beta portfolio is a portfolio whose beta with respect to the market portfolio is equal to zero:

$$\beta_Z = \frac{\text{Cov}(R_Z, R_M)}{\text{Var}(R_M)} = 0. \quad (7)$$

Therefore,

$$\boxed{\text{Cov}(R_Z, R_M) = 0.} \quad (8)$$

An important distinction is that

$$\boxed{\text{zero beta} \neq \text{risk free.}} \quad (9)$$

The zero-beta portfolio may still have positive variance:

$$\text{Var}(R_Z) > 0. \quad (10)$$

Thus, the zero-beta portfolio may contain idiosyncratic risk, but it has no systematic exposure to movements in the market portfolio.

5 Why Does $E(R_Z)$ Replace R_f ?

Consider the standard CAPM:

$$E(R_i) = R_f + \beta_i [E(R_M) - R_f]. \quad (11)$$

Setting

$$\beta_i = 0$$

gives

$$E(R_i) = R_f. \quad (12)$$

Therefore, under the standard CAPM, the intercept of the Security Market Line is the risk-free rate.

When risk-free borrowing and lending are unavailable, the relevant portfolio with zero market exposure is instead the zero-beta portfolio.

Therefore,

$$E(R_i) = E(R_Z) + \beta_i [E(R_M) - E(R_Z)]. \quad (13)$$

At $\beta_i = 0$,

$$E(R_i) = E(R_Z). \quad (14)$$

Hence, $E(R_Z)$ replaces R_f as the intercept of the Security Market Line.

6 Comparison with the Sharpe–Lintner CAPM

The standard CAPM is

$$\boxed{E(R_i) = R_f + \beta_i [E(R_M) - R_f]}. \quad (15)$$

The Black zero-beta CAPM is

$$\boxed{E(R_i) = E(R_Z) + \beta_i [E(R_M) - E(R_Z)]}. \quad (16)$$

Both pricing relationships pass through the market portfolio:

$$\beta_M = 1, \quad (17)$$

and therefore

$$E(R_i) = E(R_M) \quad \text{when} \quad \beta_i = 1. \quad (18)$$

However, under restricted borrowing, the zero-beta expected return may exceed the risk-free lending rate:

$$R_f < E(R_Z) < E(R_M). \quad (19)$$

As a consequence, the Black Security Market Line has

- a higher intercept, and
- a smaller slope.

The slope of the standard CAPM is

$$E(R_M) - R_f, \quad (20)$$

whereas the slope of the Black zero-beta CAPM is

$$E(R_M) - E(R_Z). \quad (21)$$

Since

$$E(R_Z) > R_f,$$

we have

$$E(R_M) - E(R_Z) < E(R_M) - R_f. \quad (22)$$

Thus, the Black Security Market Line is flatter.

7 Economic Intuition for the Flatter Security Market Line

Suppose investors can borrow freely at R_f .

An aggressive investor who wants more systematic risk does not need to buy individual high-beta stocks. Instead, the investor can simply borrow money and buy more of the market portfolio.

For example,

$$150\% \text{ Market} - 50\% \text{ Risk-Free Asset} \quad (23)$$

produces approximately

$$\beta = 1.5. \quad (24)$$

Now suppose borrowing at the risk-free rate is prohibited.

Investors who want higher systematic risk can no longer simply lever the market portfolio. As a result, they have greater demand for high-beta securities.

Greater demand for high-beta securities increases their prices:

$$P_{\text{high-}\beta} \uparrow. \quad (25)$$

Higher prices imply lower expected returns:

$$E(R_{\text{high-}\beta}) \downarrow . \quad (26)$$

The resulting equilibrium therefore produces a flatter relation between beta and expected return.

Relative to the standard CAPM,

$$\boxed{\text{Low-beta assets earn relatively higher expected returns}} \quad (27)$$

and

$$\boxed{\text{High-beta assets earn relatively lower expected returns.}} \quad (28)$$

This is one of the most important economic implications of Black's model.

8 Two-Fund Separation

Another important result is that two-fund separation continues to hold even without unrestricted borrowing and lending at the risk-free rate.

Efficient portfolios can be represented as combinations of

1. the market portfolio M , and
2. the zero-beta portfolio Z .

For an efficient portfolio p ,

$$R_p = \beta_p R_M + (1 - \beta_p) R_Z. \quad (29)$$

Taking expectations gives

$$E(R_p) = \beta_p E(R_M) + (1 - \beta_p) E(R_Z) \quad (30)$$

$$= E(R_Z) + \beta_p [E(R_M) - E(R_Z)]. \quad (31)$$

Thus,

$$\boxed{E(R_p) = E(R_Z) + \beta_p [E(R_M) - E(R_Z)] .} \quad (32)$$

This shows that the linear beta-pricing relationship follows naturally from the two-fund representation.

9 Empirical Interpretation

The standard Sharpe–Lintner CAPM can be written as

$$E(R_i) = \gamma_0 + \gamma_1 \beta_i, \quad (33)$$

where the model imposes

$$\gamma_0 = R_f \quad (34)$$

and

$$\gamma_1 = E(R_M) - R_f. \quad (35)$$

The Black zero-beta CAPM instead implies

$$\boxed{\gamma_0 = E(R_Z)} \quad (36)$$

and

$$\boxed{\gamma_1 = E(R_M) - E(R_Z)}. \quad (37)$$

Therefore, Black’s model does not require the intercept in a cross-sectional regression of expected returns on beta to equal the risk-free rate.

In particular, the model can accommodate

$$\gamma_0 > R_f \quad (38)$$

together with

$$\gamma_1 < E(R_M) - R_f. \quad (39)$$

This corresponds to an empirically flatter Security Market Line.

10 Connection to Early Empirical CAPM Evidence

Early empirical work on the CAPM found a pattern in which

- low-beta portfolios tended to earn higher returns than predicted by the Sharpe–Lintner CAPM, while
- high-beta portfolios tended to earn lower returns than predicted.

In other words, the empirical relation between expected returns and beta appeared approximately linear, but flatter than the standard CAPM predicted.

The Black zero-beta CAPM provides a theoretical explanation for this pattern.

The empirical relationship can take the form

$$E(R_i) = \gamma_0 + \gamma_1 \beta_i, \quad (40)$$

with

$$\gamma_0 > R_f \quad (41)$$

and

$$\gamma_1 < E(R_M) - R_f. \quad (42)$$

Therefore, evidence of a Security Market Line that is too flat does not necessarily imply that the basic beta-pricing logic is completely wrong.

Instead, it may indicate that the assumption of unlimited borrowing at the risk-free rate is unrealistic.

11 Main Difference Between the Two CAPM Versions

The central comparison can be summarized as follows:

	Sharpe–Lintner CAPM	Black Zero-Beta CAPM
Intercept	R_f	$E(R_Z)$
Slope	$E(R_M) - R_f$	$E(R_M) - E(R_Z)$
Zero-beta return	R_f	$E(R_Z)$
Risk-free borrowing	Required	Not required
Expected return–beta relation	Linear	Linear

12 The Key Equation to Remember

If only one equation is remembered from Black’s model, it should be

$$E(R_i) = E(R_Z) + \beta_i [E(R_M) - E(R_Z)]. \quad (43)$$

Compared with the Sharpe–Lintner CAPM,

$$E(R_i) = R_f + \beta_i [E(R_M) - R_f], \quad (44)$$

the essential modification is

$$R_f \longrightarrow E(R_Z). \quad (45)$$

The economic mechanism can be summarized as

$$\text{Restricted Borrowing} \implies \text{Higher SML Intercept} \implies \text{Flatter SML}. \quad (46)$$

13 Key Takeaways

- Black shows that unrestricted borrowing at the risk-free rate is **not necessary** for a linear relation between expected returns and market beta.
- When risk-free borrowing is restricted, the expected return of a **zero-beta portfolio** replaces the risk-free rate as the intercept of the Security Market Line.
- A zero-beta portfolio is not necessarily risk-free. It satisfies

$$\text{Cov}(R_Z, R_M) = 0,$$

but may still have positive variance.

- The Black Security Market Line is generally flatter than the Sharpe–Lintner Security Market Line when

$$E(R_Z) > R_f.$$

- Restricted borrowing increases investors' demand for high-beta securities, raising their prices and reducing their expected returns.
- The model therefore provides a possible explanation for the empirical tendency of low-beta securities to outperform and high-beta securities to underperform relative to the predictions of the standard CAPM.
- Black does **not** reject beta pricing. Instead, he modifies the equilibrium intercept while preserving the linear beta-pricing relation.

14 One-Sentence Summary

Black's zero-beta CAPM shows that even when investors cannot borrow freely at the risk-free rate, equilibrium expected returns remain linear in market beta, with the expected return on a zero-beta portfolio replacing the risk-free rate as the intercept of the Security Market Line.